



FACULTY OF INFORMATICS & DESIGN

Diploma in Information & Communication Technology

PROJECT 3: PRT362S

T2 Individual Report - Deliverable 2

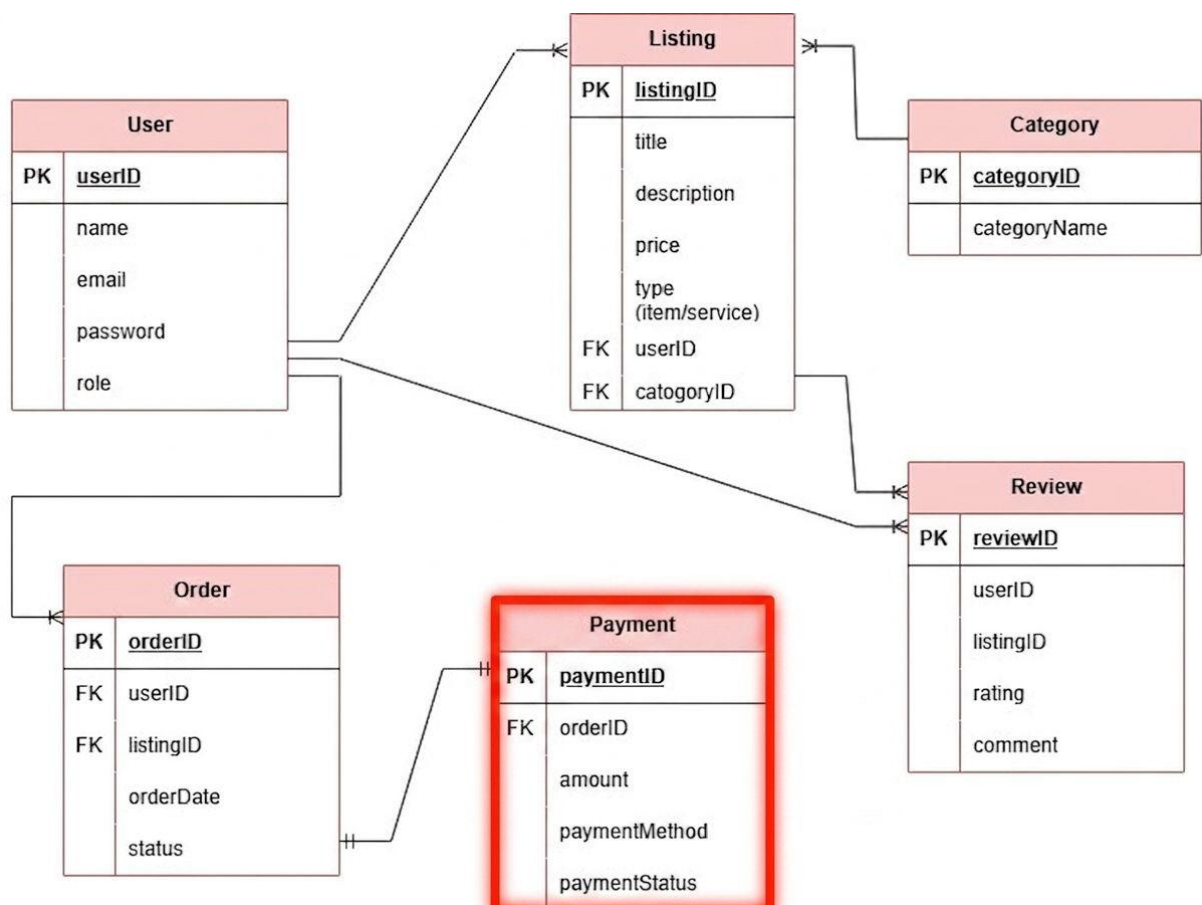
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Student Number	230211216
Course Code	PRT362S
Group Number	WO2
System Component	Payment Entity / paymentID
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Overview

This submission will highlight my individual contribution to this group assignment. My team and I are creating a digital CPUT marketplace, which would be a digital marketplace for CPUT university students across all campuses. The marketplace allows users to trade items such as physical textbooks, stationery, and the advertising of tutoring services, etc. This digital centralised marketplace is called 'CommonCents'.

In this project, I am the System Analyst. I am responsible for analysing the problem and defining the system requirements. This includes identifying how the marketplace should function, what features are needed, and ensuring that the system design aligns with the project objectives. I am also responsible for the Payment page/gate of the website. I have demonstrated this in my ERD diagram below.

ERD DIAGRAM (PAYMENT ENTITY)



The ERD illustrates the core relationships among the system's primary entities: **User**, **Listing**, **Category**, **Order**, **Payment**, and **Review**. My specific focus is the **Payment** entity, where my responsibilities centre on managing the **paymentID** and its integration within the overall order process.

Payment Entity: ERD Relationships

- **paymentID**: unique identifier reference or receipt for each transaction in order to make tracking of payments easy to identify in the database/system
- **orderId**: a reference that matches the payment to the exact purchase the customer made
- **amount**: the exact total amount of money being billed for the transaction
- **paymentMethod**: the method the customer uses to pay, e.g. credit card, debit card, or PayShap
- **paymentStatus**: a record that displays the current state of the transaction (e.g. successful, failed, or pending)

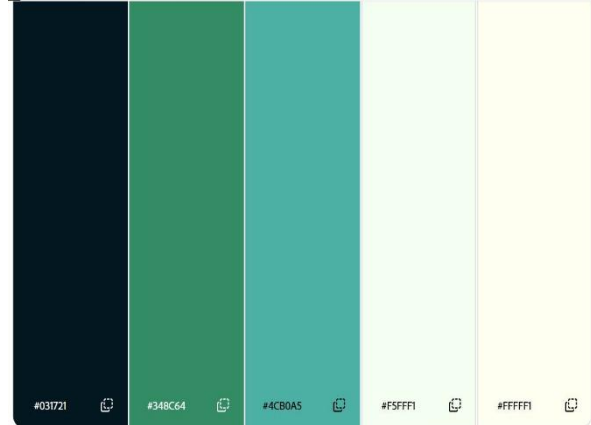
RULES FOR THE PAYMENT ENTITY

RULE	Business Rule for Payment Entity / paymentID
BR-01	Each payment transaction must have a unique paymentID.
BR-02	The system must record the exact total amount of money collected for the transaction.
BR-03	A single order can only have one successful payment.
BR-04	No payments can be made for cancelled orders.
BR-05	Customers must be able to easily view their payment history and statuses.
BR-06	You cannot edit a successful payment.
BR-07	All payment records must be kept, even in the case of refunds or order cancellations.
BR-08	Every payment must show its payment status(pending, successful, or failed)
BR-09	Payments must used an approved method for transactions. (from the given list of options)
BR-10	Every Payment must be linked to an order.

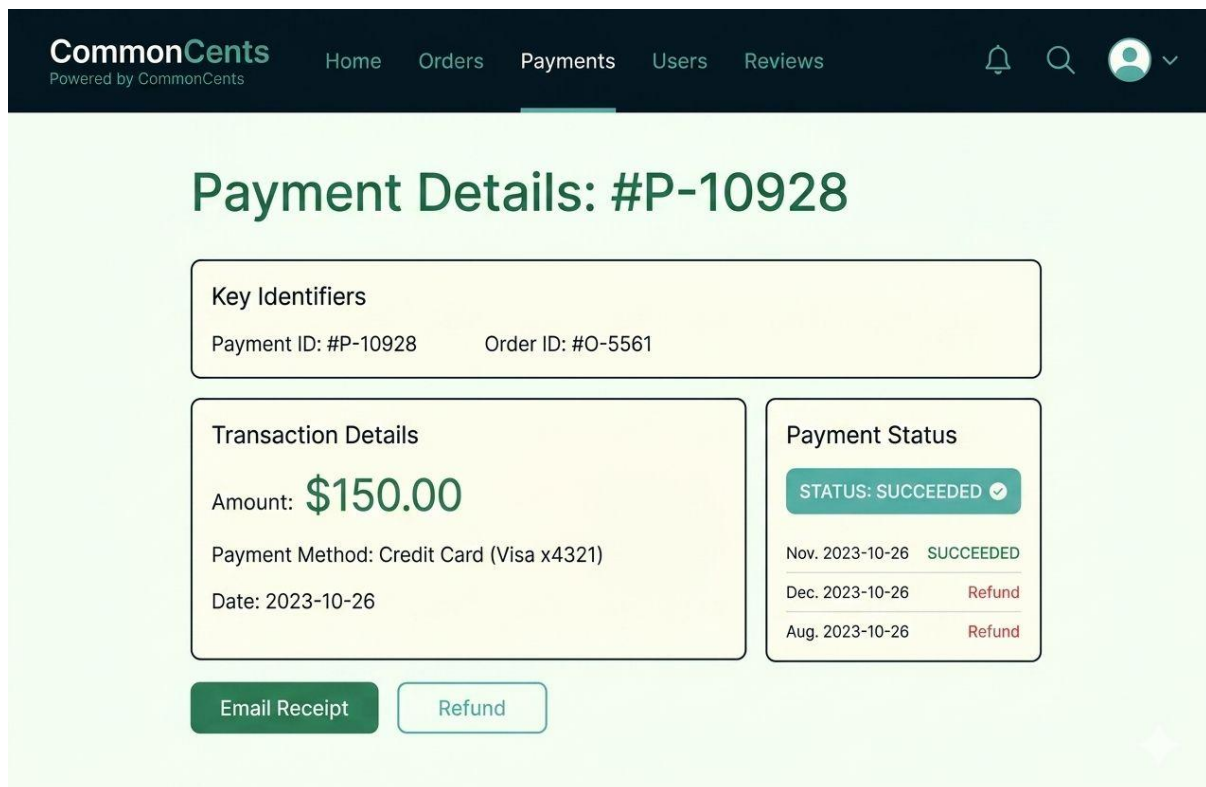
Wireframes

As a team, we had decided on these 5 colours:

- #031721
- #348C64
- #4CB0A5
- #F5FFF1
- #FFFFFF



We agreed to use these colours because they all coincide with each other in a clean professional way. Is it user-friendly in an aesthetically pleasing fashion. The dark text on soft off-white backgrounds makes the payment details easy to read without straining the eyes. Additionally, we used the bold greens to highlight important buttons and successful transactions, which keeps the design aesthetically pleasing while making it very easy for users to navigate.



Reflection

This project was a valuable opportunity to apply my skills as a System Analyst to a highly practical scenario by developing the 'CommonCents' marketplace. Focusing deeply on the Payment entity taught me how to take strict database relationships and translate them into humanized, accessible business rules.

Furthermore, working with my team members makes me realise how important teamwork is. I recognise the values in every member's individual contribution to the team- especially when we can focus on each member's skill, either as a systems analyst, a UX/UI designer or even a Project manager. Working together to implement our chosen colour palette ensured the payment interfaces were both highly functional and aesthetically pleasing for our users.